



CL-2001 Data Structures Lab # 6

Objectives:

1. Queue (Array based and Linked List based Implementation)
2. Double Ended Queue

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. There must be a block of comments at start of every question's code by students; the block should contain brief description about functionality of code.
2. Comment on every function about its functionality.
3. Use understandable name of variables.
4. Proper indentation of code is essential.
5. Make separate .cpp files for all tasks and use this format **22F-1234_Task1.cpp**.
6. First think about statement problems and then write/draw your logic on copy.
7. After copy pencil work, code the problem statement on C++ compiler.
8. Make a Microsoft Word file and paste all of your C++ code with all possible screenshots of every **task output in MS word and submit .cpp file with word file**.
9. Please submit your file in this format **22F-1234_L1**.
10. Do not submit your assignment **after the deadline**.
- 11. Do not copy code from any source otherwise you will be penalized with negative marks.**



Problem 1:

Write a C++ program to implement a queue using array with the following operations:

- **Enqueue:** Add an element to the end of the queue.
- **Dequeue:** Remove an element from the front of the queue.
- **Front:** Return the element at the front of the queue without removing it.
- **IsEmpty:** Return true if the queue is empty, false otherwise.
- **IsFull:** Return true if the queue is full, false otherwise (assuming a fixed-size array).
- **Display:** Print all elements in the queue.

Your program should also handle situations where enqueue or dequeue operations are called on an empty or full queue respectively.

Problem 2:

You are tasked with creating an online ticket booking system for a popular amusement park. The park has several rides, and visitors can book tickets for each ride. The system should ensure fair access to the rides based on a first-come, first-served basis.

Implement a queue using a linked list to manage the ticket booking system. The queue should support the following operations:

- **Enqueue:** Add a visitor to the queue when they book a ticket for a ride.
- **Dequeue:** Remove a visitor from the queue when they have enjoyed the ride.
- **Front:** Retrieve the details of the visitor at the front of the queue (the next person in line).
- **IsEmpty:** Check if the queue is empty.
- **Display:** Display the list of visitors in the queue.

Consider the following scenario:

The amusement park has three popular rides: Roller Coaster, Ferris Wheel, and Water Slide. Visitors can book tickets for any of these rides. As visitors arrive, they join the queue for their chosen ride. Once a visitor has enjoyed the ride, they are removed from the queue.

Write a C++ program that simulates this scenario using a linked list-based queue.

Note:

- Each visitor can only book one ticket for one ride at a time.
- Feel free to adapt or expand upon this scenario as needed!



Problem 3:

Write a C++ program to implement a double-ended queue (deque) with the following operations:

- **Push Front:** Add an element to the front of the deque.
- **Push Back:** Add an element to the back of the deque.
- **Pop Front:** Remove an element from the front of the deque.
- **Pop Back:** Remove an element from the back of the deque.

